



Multimodal Property Price Prediction Using Satellite Imagery and Tabular Data

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Project Summary

This research investigates a novel multimodal machine learning approach for property price prediction by integrating high-resolution satellite imagery with traditional tabular real estate data. We develop a deep learning framework that combines convolutional neural networks (CNNs) for visual feature extraction from Sentinel Hub satellite images with gradient boosting models (XGBoost) for structured data processing. The fusion architecture leverages spatial environmental features: including vegetation density, urban infrastructure patterns, and geographic context alongside conventional property attributes such as square footage, location, and amenities. Our results demonstrate that incorporating visual satellite data significantly enhances prediction accuracy compared to tabular-only models, revealing previously underutilized environmental factors that influence property valuations. This work contributes to the growing field of geospatial machine learning and offers practical applications for real estate analytics and urban planning.

Dataset & Data Collection



Tabular Data Sources

Comprehensive property records including square footage, bedrooms, bathrooms, year built, lot size, property type, and historical transaction prices.



Satellite Imagery Acquisition

High-resolution RGB and near-infrared imagery from Sentinel Hub API, capturing pixel regions centered on each property location with 10-meter spatial resolution.



Preprocessing Pipeline

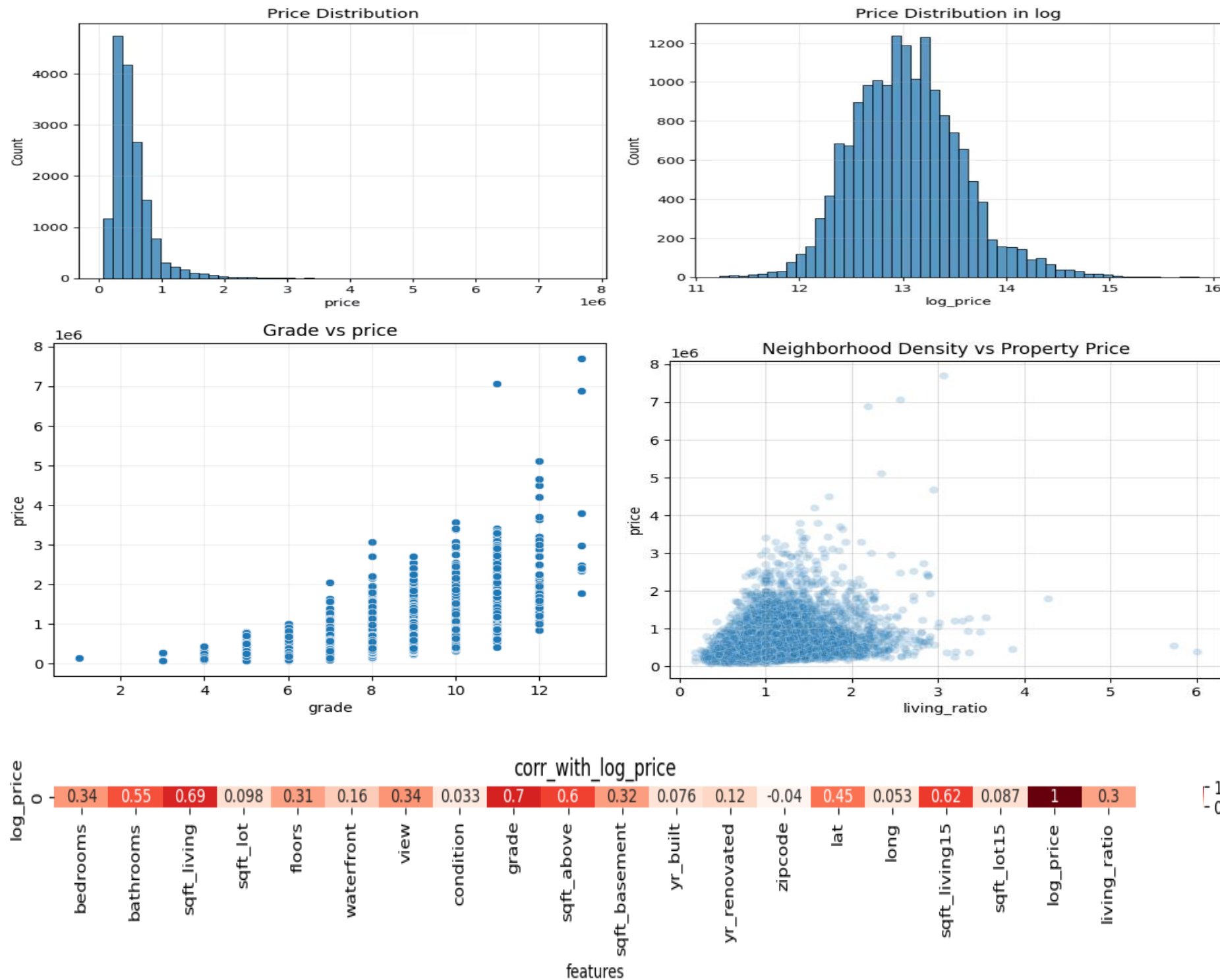
Standardized image normalization, cloud filtering, temporal averaging across multiple captures, and synchronized alignment with property coordinates ensuring data quality.



Exploratory Data Analysis

Key Insights from EDA

- **Price distribution:** Right-skewed with median at \$450K, requiring log transformation for modeling
- **Grade and Price:** Higher grade results in higher prices showing positive relation
- **Neighborhood Density vs Property Price:** moderate living ratio has high property value with extreme densities
- **Feature correlations:** Strong positive relationships between square footage (r=0.69), grade (r=0.7) and price



Visual Feature Analysis: Environmental Factors

Gradient-weighted Class Activation Mapping (Grad-CAM) reveals which visual features the CNN emphasizes when predicting property values, providing interpretability for the image-based model.

Images and dimension which have high correlation with prices



Grad-CAM: Image no 2659 and Regions Influencing Prediction

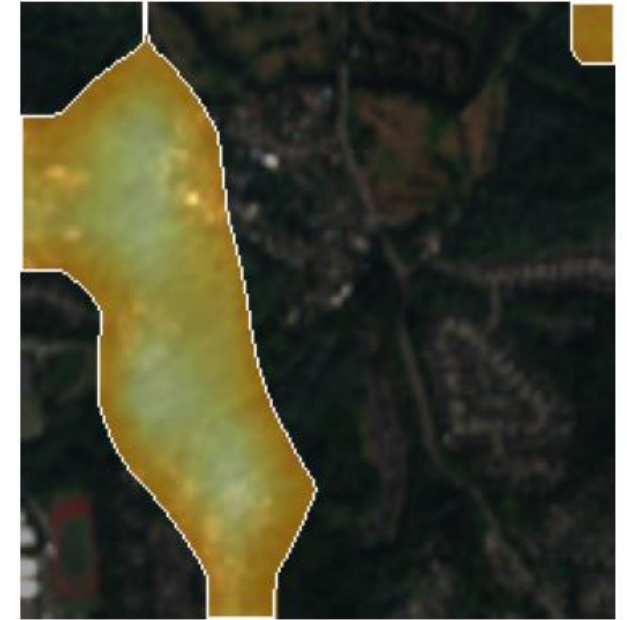


Grad-CAM highlights strong activations along coastal boundaries and dense built-up regions, indicating that proximity to water and urban infrastructure significantly influence property prices

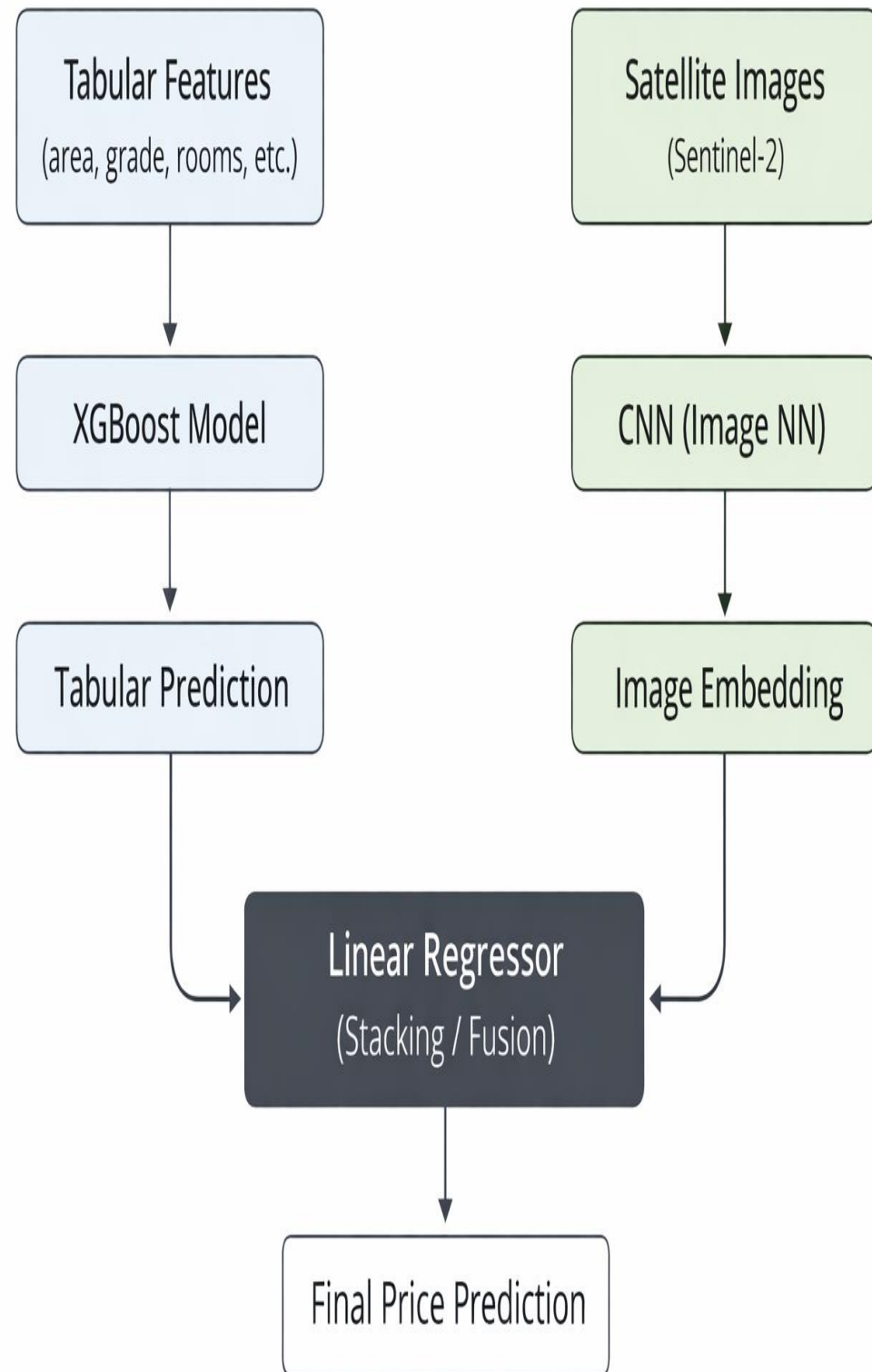
Images and dimension which have low correlation with prices



Grad-CAM: Image no 4676 and Regions Influencing Prediction



The model shows weaker and more diffuse attention over vegetated and sparsely developed areas, suggesting these visual features contribute less to price prediction



Model Architecture Design

01

Image Feature Extraction

ResNet-50 CNN pretrained on ImageNet, fine-tuned on satellite imagery to extract 2048-dimensional visual embeddings and then converted to 128-dimensional visual embeddings using PCA

02

Tabular Feature Processing

XGBoost gradient boosting with hyperparameter optimization handling structured property attributes and location data

03

Feature Fusion Layer

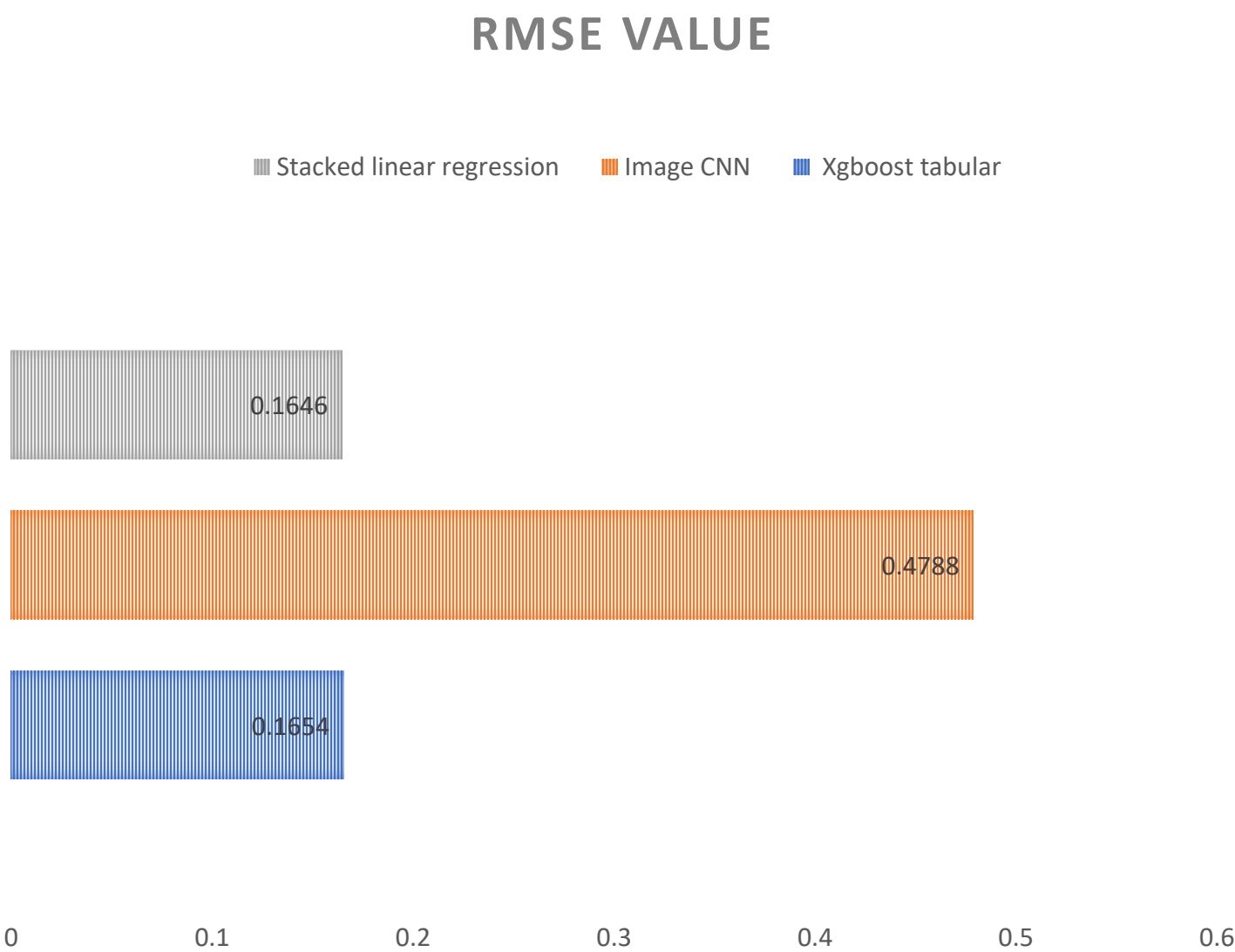
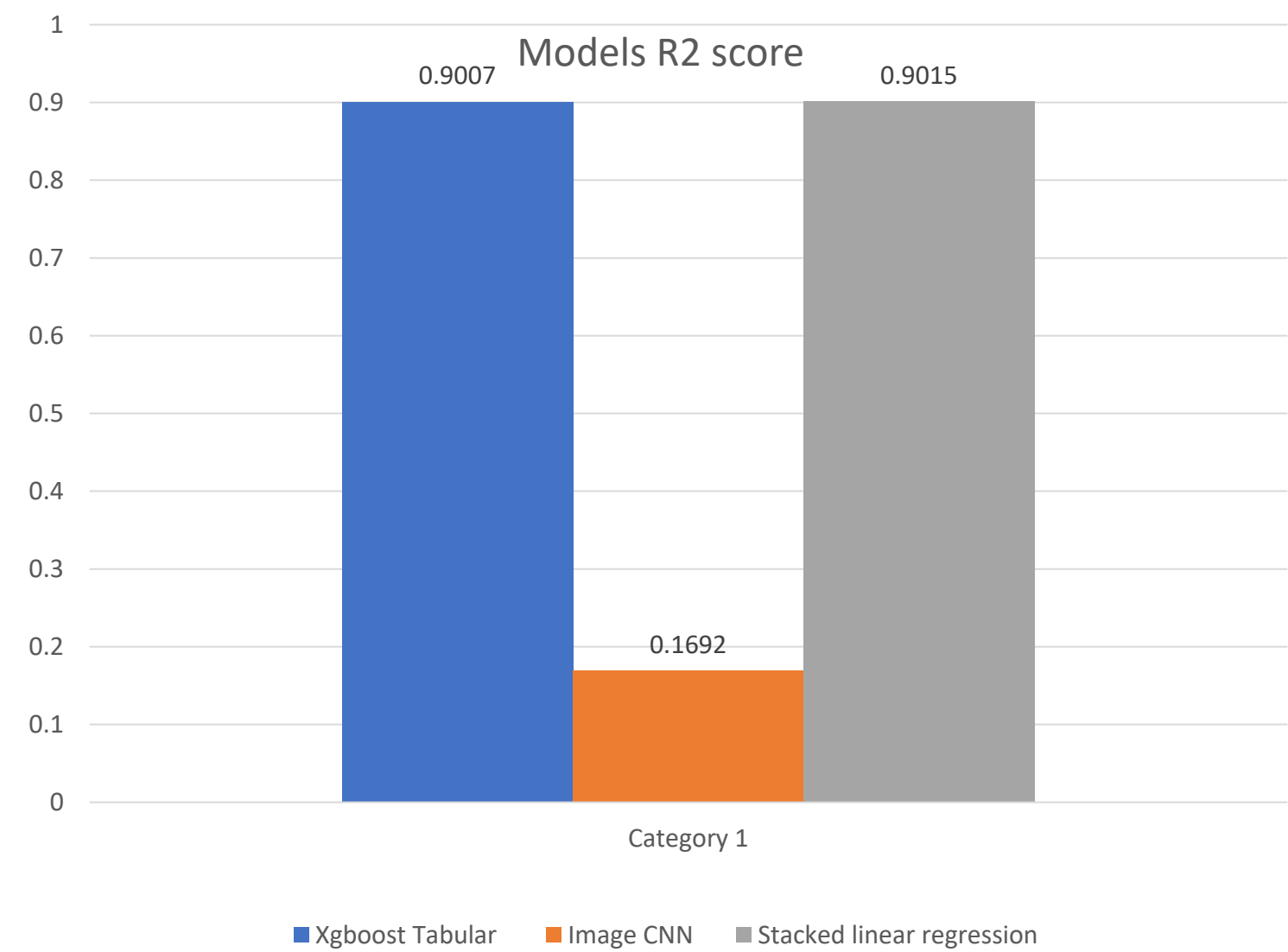
The feature fusion layer combines predictions from the tabular XGBoost model and embeddings from the image CNN using a linear regressor, enabling the model to jointly leverage structured attributes and visual environmental context for improved price estimation

04

Regression Output

The final regression layer produces the property price prediction using mean squared error (MSE) loss, with the model trained end-to-end while applying different learning rates to the tabular and image branches for stable and efficient optimization.

Model Performance & Results



- The tabular XGBoost model achieves performance with an R² of ~0.90 and low RMSE, indicating strong predictive power from structured features
- The image-only CNN performs significantly worse, highlighting the difficulty of predicting prices using visual information alone
- The stacked linear regression model effectively integrates both modalities, maintaining the highest R² and lowest RMSE, demonstrating that visual features provide complementary context when combined with tabular data.